Mechanistic Interpretability of LLM Reasoning via Knowledge Manipulation
Final Report

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1 Introduction

Large language models (LLMs) have demonstrated strong performance on a wide range of reasoning tasks [32, 3, 24], including multi-step question answering, symbolic inference, and rule-based decision making [4, 22]. Their success has raised an important question: how do these models internally perform reasoning? Although recent work has shown that LLMs can generate impressive reasoning outputs, most evidence remains behavioral, typically based on final-answer accuracy or the quality of generated reasoning traces [13, 34, 28]. Such results indicate that reasoning-like behavior is present, but they do not by themselves explain how reasoning is implemented inside the model.

This limitation has made interpretability an increasingly important direction for reasoning research. In particular, mechanistic interpretability seeks to explain model behavior in terms of internal computational components and their functional interactions [8, 6, 14, 30, 16]. Rather than treating LLMs as black boxes, it aims to identify how layers, attention heads, neurons, or larger circuits contribute to specific behaviors. For reasoning tasks, this perspective is especially valuable because successful performance depends not only on producing the correct answer, but also on how relevant information is retrieved, maintained, transformed, and combined across intermediate computational steps.

However, current approaches to reasoning interpretability still face a central challenge. Many existing studies rely on observational evidence, such as attention patterns, attribution scores, or probing-based signals [29, 19, 33]. Although these methods can reveal useful correlations, they often provide limited evidence about functional necessity. An internal feature may co-occur with successful reasoning without being causally responsible for it. As a result, a more intervention-based framework is needed in order to move from descriptive analysis toward mechanistic explanation.

This paper addresses this challenge through knowledge manipulation. The core idea is to treat knowledge not merely as a static capability stored in the model, but as an experimental variable for probing reasoning mechanisms. By introducing controlled changes to the knowledge conditions underlying a reasoning task, it becomes possible to observe how these changes influence both model outputs and internal computations. If a particular internal component genuinely supports knowledge-dependent reasoning, then targeted modifications to relevant knowledge should induce systematic changes in that component's behavior. In this way, knowledge manipulation functions as a causal probe for identifying how reasoning is realized inside the model.

Building on this perspective, the present study investigates how controlled changes in knowledge conditions affect internal reasoning processes in LLMs. Rather than focusing only on final-answer correctness, the study traces how manipulated knowledge propagates through internal representations and contributes to reasoning outcomes. This provides a more mechanistically grounded perspective on LLM reasoning and connects reasoning research with intervention-based interpretability.

2 Background

2.1 Reasoning in Large Language Models

Reasoning has become one of the most actively studied capabilities of LLMs. A large body of work has shown that these models can solve tasks involving multi-step inference, compositional question answering, mathematical reasoning, and rule application [12, 35, 22]. In many cases, their outputs suggest an ability to process intermediate information and produce structured conclusions rather than simple pattern completion.

At the same time, the interpretation of such behavior remains contested. Strong task performance does not necessarily imply that the model is reasoning in a robust or human-like sense. More importantly, even when a model succeeds on a reasoning benchmark, the internal basis of that success often remains unclear [2, 20]. A correct answer may result from shallow shortcuts, memorized correlations, partial pattern matching, or genuinely knowledge-dependent inference [26, 36]. Behavioral evaluation alone is often insufficient to distinguish among these possibilities [11, 37].

For this reason, understanding reasoning in LLMs requires not only measuring outputs but also analyzing the internal computational processes that generate them. This shift from behavioral performance to internal mechanism is central to the present study.

2.2 Mechanistic Interpretability

Mechanistic interpretability provides a framework for analyzing model behavior at the level of internal computation [21, 1]. Its goal is to explain how specific components of a neural model contribute to the transformations that ultimately produce an output. Existing work in this area has developed methods such as activation analysis, causal tracing, activation patching, and circuit discovery, all of which aim to uncover the computational structure underlying model behavior [5, 10, 14].

This framework is particularly relevant for reasoning because reasoning is inherently process-oriented. A model solving a reasoning task must not only access relevant information, but also preserve and transform that information across multiple stages of computation [17, 27, 23]. Mechanistic interpretability makes it possible to investigate where such operations occur, how they are distributed across layers or modules, and which components are functionally important.

Nevertheless, one methodological difficulty remains prominent. Much interpretability evidence is correlational rather than causal. A component may appear important because it is active during successful reasoning, yet its activity alone does not prove that it plays a necessary role [10, 5]. To make stronger mechanistic claims, interpretability analysis benefits from controlled interventions that alter the conditions of reasoning and reveal how internal computations respond.

2.3 Knowledge Manipulation as an Interpretability Tool

A promising form of controlled intervention is knowledge manipulation. Many reasoning tasks depend on the use of relevant knowledge, whether in the form of facts, premises, rules, or structured conditions [9, 7, 25]. If this knowledge is changed in a targeted way, the resulting differences can reveal how reasoning depends on it internally.

Knowledge manipulation is often studied in the context of knowledge editing or factual robustness [31, 14, 15, 18]. In those settings, the main question is whether the model can be updated or whether its answers remain stable under changed facts. The present study takes a different perspective. Here, knowledge manipulation is not the final objective, but a tool for understanding reasoning mechanisms. By systematically altering knowledge conditions, one can examine which internal representations are sensitive to those changes, where the effect emerges in the computation, and whether the affected components play a functional role in producing the final answer.

This perspective is useful because it links two dimensions that are often studied separately: knowledge and reasoning. In many tasks, reasoning is not independent of knowledge; it is a computation over knowledge-bearing representations. Manipulating knowledge therefore provides a principled way to probe reasoning from the inside. If a model's internal reasoning process genuinely depends on a given premise or fact, then modifying that knowledge should lead to structured and traceable changes in internal states. Such changes can provide stronger evidence for mechanism than observation alone.

3 Research Questions

Based on the above motivation, this study focuses on the following research questions:

- **RQ1:** Does controlled manipulation of task-relevant knowledge systematically affect the final reasoning outputs of LLMs?
- **RQ2:** At which layers or internal components do knowledge manipulations induce the largest representation changes during reasoning?
- **RQ3:** Can causal intervention on these knowledge-sensitive internal states recover or alter reasoning behavior in predictable ways?

These questions are designed to move from behavioral sensitivity to internal localization and finally to causal validation.

4 Methods

4.1 Study Objective

The objective of this study is to investigate the internal mechanisms of knowledge-dependent reasoning in LLMs through controlled knowledge manipulation. The analysis focuses on how targeted changes to relevant knowledge conditions affect model behavior at both the output level and the internal representation level. More specifically, the study seeks to identify which internal components are involved in retrieving, maintaining, or transforming knowledge during reasoning.

4.2 Task Design

The study focuses on reasoning tasks in which knowledge dependence is explicit and intervention points can be clearly defined. Appropriate task settings include structured multi-step reasoning, multi-hop question answering, and rule-based inference, where a correct answer depends on identifiable facts, premises, or conditions. These tasks are particularly suitable because they allow the construction of controlled example variants that differ only in a targeted knowledge-related dimension.

For each original example, manipulated counterparts are created by modifying one meaningful aspect of the knowledge condition while preserving the overall task structure. Such manipulations may include factual substitution, premise perturbation, rule-condition modification, conflict injection, or the insertion of distracting information. The purpose of this design is to isolate the effect of knowledge changes while minimizing unrelated variation across examples.

This paired construction allows the study to compare not only whether the answer changes, but also how the internal reasoning process changes when the knowledge basis of the problem is altered.

4.3 Knowledge Manipulation Design

The effectiveness of this framework depends on the design of the manipulations themselves. Each manipulation must be targeted, interpretable, and semantically controlled. The goal is not to create arbitrary adversarial noise, but to alter a specific knowledge-bearing element that is relevant to the reasoning process.

For this reason, manipulations are designed to preserve as much of the original problem structure as possible while changing one meaningful premise, fact, or rule. This makes it easier to attribute observed differences in internal states to the manipulated knowledge condition rather than to incidental variation in wording or task form. It also makes the resulting comparisons more suitable for mechanistic interpretation.

Under this design, manipulated examples function as probes for the model's internal reasoning process. They create controlled opportunities to test whether particular internal signals track, encode, or use the modified knowledge during inference.

4.4 Mechanistic Analysis

The core analysis examines how knowledge manipulation affects internal model states. Several interpretability methods are used for this purpose, including hidden-state comparison across original and manipulated conditions, layer-wise sensitivity analysis, and causal intervention methods such as activation patching. These tools make it possible to identify which parts of the model are most responsive to changes in knowledge and whether those parts contribute functionally to the final prediction.

The analysis is designed to move beyond output-level perturbation testing. Instead of asking only whether the answer changes, it asks where the internal computation changes, how that change propagates through the network, and whether altering a given internal signal changes the resulting reasoning behavior. This provides a stronger basis for mechanistic claims than descriptive comparison alone.

In particular, if restoring an internal activation from the original condition to the manipulated condition also restores the original reasoning behavior, that would provide evidence that the corresponding component plays a causal role in mediating the use of knowledge during reasoning. Conversely, if a component shows large observational differences but patching it has little effect, its functional importance may be limited.

4.5 Evaluation

Evaluation is conducted at multiple levels. Final-answer behavior provides one indicator of the model's sensitivity to manipulated knowledge, but it is not sufficient on its own. The study also evaluates differences in internal representations between original and manipulated conditions, the degree to which the effects of manipulation can be localized to particular layers or components, and whether causal interventions on those components alter reasoning behavior in predictable ways.

This multi-level evaluation framework is intended to capture not only whether reasoning changes, but also how it changes internally. In doing so, it supports a more precise account of the relationship between knowledge and reasoning in LLMs.

5 Experiment Design

5.1 Experimental Setting

To make the study concrete, the experiment was conducted on a knowledge-dependent rule-based reasoning task using the ProofWriter [25] benchmark. The experiment uses paired examples consisting of an *original version* and a *manipulated version*. The two versions share the same question and most of the theory context, but differ in one key supporting fact that is necessary for deriving the original answer.

In this project, the experimental data consisted of 4096 paired examples. The pairs were constructed automatically from ProofWriter theories in the open-world assumption (OWA) setting. For each original example, a manipulated counterpart was created by removing one supporting fact that appeared in the proof of a positively provable question, while keeping the question unchanged. The manipulated label was then recomputed using a simple forward-chaining procedure over the remaining facts and rules. This construction makes the intervention relatively clean: the reasoning target remains fixed, while the supporting knowledge condition is altered in a targeted way.

5.2 Example of Knowledge Manipulation

An example of the manipulation strategy is shown below:

> **Original:** "The cow is big. If something is big, then it chases the dog. If the cow chases the dog, then the cow sees the rabbit. Question: Does the cow see the rabbit?"
>
> **Manipulated:** "If something is big, then it chases the dog. If the cow chases the dog, then the cow sees the rabbit. Question: Does the cow see the rabbit?"

In the manipulated version, a supporting fact required by the original proof is removed, while the question and the rest of the rule structure are preserved. This reflects the core intervention used in the experiment: rather than rewriting the entire reasoning problem, the manipulation changes a localized piece of task-relevant knowledge. If the model's reasoning genuinely depends on that knowledge, both the final output and internal representations should respond systematically to the intervention.

5.3 Model and Extraction Procedure

The model used in this study was `mistralai/Mistral-7B-Instruct-v0.3`. For each example, the model was prompted using the same answer format under both original and manipulated conditions. Hidden states were extracted from all transformer layers during inference, and layer-wise representation differences were measured using the hidden state of the final input position.

For each example pair, the following information was recorded:

- the model's final answer under the original condition;

- the model's final answer under the manipulated condition;
- hidden-state representations across layers for both conditions;
- activation patching results for the most knowledge-sensitive layers.

After representation-level analysis, the three layers with the highest mean divergence between original and manipulated conditions were selected as candidate patching layers, namely layers 19, 20, and 21.

5.4 Metrics

Three types of metrics were used:

Output-level metric. The first metric is answer behavior under knowledge manipulation. Specifically, the study measures original-condition accuracy, manipulated-condition accuracy, the rate at which the model changes its answer after manipulation, and the correct update rate, i.e., the proportion of cases in which the model changes its answer and the manipulated answer matches the recomputed manipulated label.

Representation-level metric. The second metric is hidden-state divergence between the original and manipulated conditions. In this experiment, divergence was measured by cosine distance layer by layer. Larger divergence at a given layer suggests stronger sensitivity of that layer to the manipulated knowledge.

Causal metric. The third metric is patching effectiveness. If patching activations from the original condition into the manipulated condition partially restores the original answer, then the patched layer is likely to play a causal role in mediating the use of knowledge during reasoning.

5.5 Hypotheses

The experiment was guided by the following hypotheses:

- **H1:** Knowledge manipulation will change model behavior, although the magnitude of the change may be limited because the model does not always rely fully on the removed supporting fact.
- **H2:** Internal representation differences will be concentrated in middle-to-late layers rather than being uniformly distributed across the network.
- **H3:** Activation patching on the most knowledge-sensitive layers will more strongly recover the original answer than patching less sensitive layers.

6 Results

6.1 Output-Level Results

Table 1 reports the model's output behavior under original and manipulated conditions.

Several observations can be made from these results. First, the model performs substantially better on the manipulated condition than on the original condition. This asymmetry likely reflects the construction of the manipulated data: removing a supporting fact often turns an originally

Table 1: Output-level results under knowledge manipulation.

Metric	Value	Notes
Original-condition accuracy	0.3467	Accuracy on original ProofWriter examples
Manipulated-condition accuracy	0.7664	Accuracy on manipulated examples
Answer change rate after manipulation	0.1682	Proportion of pairs with changed answers
Correct update rate	0.1406	Changed answer and matched manipulated label

provable statement into an easier `unknown`-style case, which the model handles more reliably than the original positive reasoning problem. Second, the answer change rate is 0.1682, showing that only a minority of interventions lead the model to revise its output. Third, the correct update rate is 0.1406, which is close to the overall change rate. This indicates that when the model does change its answer, the change is often in the correct direction, but many manipulated cases still fail to trigger revision at all.

Overall, the output-level results suggest that the knowledge intervention has a measurable behavioral effect, but the effect is limited. In many cases, the model appears relatively insensitive to the removal of a supporting fact, even though the logical status of the question has changed.

6.2 Representation-Level Results

To examine where knowledge manipulation affects internal computation, hidden-state differences were measured layer by layer between the original and manipulated conditions. The results are summarized in Table 2.

Table 2: Layer-wise hidden-state divergence between original and manipulated conditions.

Layer range	Mean divergence	Observation
Early layers	0.000634	Very small effect
Middle layers	0.008374	Clear increase in sensitivity
Late layers	0.009638	Highest overall sensitivity
Peak layer	21 (0.01535)	Strongest localized divergence

The representation-level results show a clear progression across the network. Early layers exhibit almost no divergence between original and manipulated inputs, suggesting that these layers mainly process shared lexical and local contextual information. By contrast, divergence increases substantially in the middle layers and remains high in the late layers. The strongest effect appears at layer 21, whose mean cosine-distance divergence reaches 0.01535.

This pattern supports the view that the impact of knowledge manipulation is not uniformly distributed across the model. Instead, it emerges primarily in a relatively narrow band of middle-to-late layers. These layers are therefore plausible candidates for mediating knowledge-dependent transformations relevant to the final reasoning output.

6.3 Causal Intervention Results

Activation patching was conducted on the three layers that showed the highest representation-level sensitivity, namely layers 19, 20, and 21. Table 3 summarizes the patching outcomes.

Table 3: Activation patching results.

Patched layer/component	Recovery rate	Observation
Layer 19	0.35	Moderate recovery to original output
Layer 20	0.40	Best recovery effect
Layer 21	0.35	Moderate recovery to original output
Best-performing patch	Layer 20 (0.40)	Highest recovery to original

The patching results provide initial causal support for the representation analysis. Patching activations from the original condition into the manipulated condition at layers 19–21 partially restores the original answer, with the strongest effect observed at layer 20. This aligns well with the layer-wise divergence results, where layers 19–21 were identified as the most knowledge-sensitive part of the network.

At the same time, the recovery effect is clearly partial rather than complete. Even the best-performing patch only restores the original answer in 40% of the patching cases. This suggests that the knowledge-dependent reasoning signal is not localized to a single layer alone. Instead, it is likely distributed across multiple nearby stages or depends on interactions that are only partially captured by the current single-layer patching method.

7 Discussion

The results provide empirical support for the main claim of this project: controlled knowledge manipulation can serve as a practical tool for studying the internal mechanisms of reasoning in LLMs.

First, the output-level findings show that removing a supporting fact does affect model behavior, but the effect is weaker than a purely logic-driven account would predict. The model changes its answer in only 16.82% of manipulated cases, suggesting that much of its behavior is not tightly coupled to the formal proof structure captured by the dataset construction. This may reflect heuristic answering, partial reliance on surface regularities, or a limited sensitivity to missing support in the prompt.

Second, the representation-level results show a more structured picture internally than the output-level results alone would suggest. The effect of manipulation is minimal in early layers, grows strongly in the middle layers, and remains high in the late layers, peaking at layer 21. This indicates that the model does register the difference between original and manipulated knowledge conditions internally, even when that difference does not always translate into an output change.

Third, the activation patching results provide a useful bridge between correlation and causation. The fact that patching layers 19–21 can partially recover the original answer, especially at layer 20, suggests that these layers are not merely co-activated with reasoning success. Rather, they appear to participate functionally in the processing of knowledge relevant to the final prediction.

Taken together, these findings suggest a layered interpretation of knowledge-dependent reasoning in this setting. The manipulated fact does leave a detectable trace in the model's internal representations, especially in middle-to-late layers, but the model's final answer is often more stable than the formal change in logical support would warrant. This gap between internal sensitivity and behavioral revision is itself an informative result: it suggests that internal knowledge-sensitive computation and final decision behavior are related, but not perfectly aligned.

8 Limitations

This study has several limitations.

First, although the experiment includes 4096 paired examples, the manipulation itself is relatively narrow. Each manipulated example is constructed by removing a single supporting fact from a ProofWriter theory. This makes the intervention clean and interpretable, but also limits the range of knowledge changes being tested.

Second, the task domain is synthetic and rule-based. ProofWriter is well suited for controlled mechanistic analysis, but it does not capture the full complexity of naturalistic reasoning, where relevant knowledge may be implicit, redundant, or distributed across multiple statements.

Third, the present analysis mainly operates at the layer level. While this is useful for locating where the effect of knowledge manipulation is strongest, it does not identify the finer-grained circuits, attention heads, or MLP subcomponents that implement the relevant computation.

Fourth, the current patching analysis only considers a small set of top-divergence layers and uses single-layer interventions. As a result, the causal evidence remains partial. A more complete account would require multi-layer patching, token-specific patching, or component-level interventions.

Finally, the manipulated-condition accuracy is much higher than the original-condition accuracy, which suggests that the two conditions differ not only in knowledge support but also in practical difficulty for the model. This asymmetry should be kept in mind when interpreting behavioral differences.

9 Conclusion

This final report explored the mechanistic interpretability of LLM reasoning through controlled knowledge manipulation. Using 4096 paired examples derived from ProofWriter and evaluated with Mistral-7B-Instruct, the study examined how removing a proof-relevant supporting fact affects model outputs, internal representations, and causal intervention outcomes.

The results show that knowledge manipulation has a limited but measurable behavioral effect, a clearer and more localized representation-level effect in middle-to-late layers, and a partially successful causal recovery effect through activation patching. In particular, layers 19–21 emerged as the most knowledge-sensitive region of the network, with layer 20 showing the strongest recovery effect under patching.

Overall, the findings support the usefulness of knowledge manipulation as an interpretability tool. Even when the model does not always revise its final answer appropriately, the intervention reveals structured internal changes and helps localize candidate computational stages involved in knowledge-dependent reasoning. Future work can build on this framework by using more diverse manipulation types, more naturalistic reasoning tasks, and finer-grained causal analysis at the level of tokens, heads, and circuits.

References

[1] Bereska, L., and Gavves, E. Mechanistic interpretability for ai safety–a review. *arXiv preprint arXiv:2404.14082* (2024).

[2] Berglund, L., Tong, M., Kaufmann, M., Balesni, M., Stickland, A. C., Korbak, T., and Evans, O. The reversal curse: Llms trained on" a is b" fail to learn" b is a". *arXiv preprint arXiv:2309.12288* (2023).

[3] Chowdhery, A., Narang, S., Devlin, J., Bosma, M., Mishra, G., Roberts, A., Barham, P., Chung, H. W., Sutton, C., Gehrmann, S., et al. Palm: Scaling language modeling with pathways. *Journal of machine learning research 24*, 240 (2023), 1–113.

[4] Clark, P., Cowhey, I., Etzioni, O., Khot, T., Sabharwal, A., Schoenick, C., and Tafjord, O. Think you have solved question answering? try arc, the ai2 reasoning challenge. *arXiv preprint arXiv:1803.05457* (2018).

[5] Conmy, A., Mavor-Parker, A., Lynch, A., Heimersheim, S., and Garriga-Alonso, A. Towards automated circuit discovery for mechanistic interpretability. *Advances in Neural Information Processing Systems 36* (2023), 16318–16352.

[6] Dai, D., Dong, L., Hao, Y., Sui, Z., Chang, B., and Wei, F. Knowledge neurons in pretrained transformers. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2022), pp. 8493–8502.

[7] Geva, M., Khashabi, D., Segal, E., Khot, T., Roth, D., and Berant, J. Did aristotle use a laptop? a question answering benchmark with implicit reasoning strategies. *Transactions of the Association for Computational Linguistics 9* (2021), 346–361.

[8] Geva, M., Schuster, R., Berant, J., and Levy, O. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (2021), pp. 5484–5495.

[9] Han, S., Schoelkopf, H., Zhao, Y., Qi, Z., Riddell, M., Zhou, W., Coady, J., Peng, D., Qiao, Y., Benson, L., et al. Folio: Natural language reasoning with first-order logic. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing* (2024), pp. 22017–22031.

[10] Heimersheim, S., and Nanda, N. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255* (2024).

[11] Inger, N. C., Elisha, Y., Shapira, B., Rokach, L., and Cohen, S. Forget what you know about llms evaluations-llms are like a chameleon. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing* (2025), pp. 21675–21688.

[12] Kojima, T., Gu, S. S., Reid, M., Matsuo, Y., and Iwasawa, Y. Large language models are zero-shot reasoners. *Advances in neural information processing systems 35* (2022), 22199–22213.

[13] Lyu, Q., Havaldar, S., Stein, A., Zhang, L., Rao, D., Wong, E., Apidianaki, M., and Callison-Burch, C. Faithful chain-of-thought reasoning. In *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)* (2023), pp. 305–329.

[14] Meng, K., Bau, D., Andonian, A., and Belinkov, Y. Locating and editing factual associations in gpt. *Advances in neural information processing systems 35* (2022), 17359–17372.

[15] Meng, K., Sharma, A. S., Andonian, A., Belinkov, Y., and Bau, D. Mass-editing memory in a transformer. *arXiv preprint arXiv:2210.07229* (2022).

[16] Nanda, N., Chan, L., Lieberum, T., Smith, J., and Steinhardt, J. Progress measures for grokking via mechanistic interpretability. *arXiv preprint arXiv:2301.05217* (2023).

[17] Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., Mann, B., Askell, A., Bai, Y., Chen, A., et al. In-context learning and induction heads. *arXiv preprint arXiv:2209.11895* (2022).

[18] Park, H., Choi, G., Kim, M., and Jo, Y. Context-robust knowledge editing for language models. In *Findings of the Association for Computational Linguistics: ACL 2025* (2025), pp. 10360–10385.

[19] Paul, D., West, R., Bosselut, A., and Faltings, B. Making reasoning matter: Measuring and improving faithfulness of chain-of-thought reasoning. In *Findings of the Association for Computational Linguistics: EMNLP 2024* (2024), pp. 15012–15032.

[20] Press, O., Zhang, M., Min, S., Schmidt, L., Smith, N. A., and Lewis, M. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023* (2023), pp. 5687–5711.

[21] Rai, D., Zhou, Y., Feng, S., Saparov, A., and Yao, Z. A practical review of mechanistic interpretability for transformer-based language models. *arXiv preprint arXiv:2407.02646* (2024).

[22] Ravichander, A., Naik, A., Rose, C., and Hovy, E. Equate: A benchmark evaluation framework for quantitative reasoning in natural language inference. In *Proceedings of the 23rd conference on computational natural language learning (CoNLL)* (2019), pp. 349–361.

[23] Ren, J., Guo, Q., Yan, H., Liu, D., Zhang, Q., Qiu, X., and Lin, D. Identifying semantic induction heads to understand in-context learning. In *Findings of the Association for Computational Linguistics: ACL 2024* (2024), pp. 6916–6932.

[24] Suzgun, M., Scales, N., Schärli, N., Gehrmann, S., Tay, Y., Chung, H. W., Chowdhery, A., Le, Q., Chi, E., Zhou, D., et al. Challenging big-bench tasks and whether chain-of-thought can solve them. In *Findings of the Association for Computational Linguistics: ACL 2023* (2023), pp. 13003–13051.

[25] Tafjord, O., Dalvi, B., and Clark, P. Proofwriter: Generating implications, proofs, and abductive statements over natural language. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021* (2021), pp. 3621–3634.

[26] Tang, R., Kong, D., Huang, L., et al. Large language models can be lazy learners: Analyze shortcuts in in-context learning. In *Findings of the association for computational linguistics: ACL 2023* (2023), pp. 4645–4657.

[27] Todd, E., Li, M. L., Sharma, A. S., Mueller, A., Wallace, B. C., and Bau, D. Function vectors in large language models. *arXiv preprint arXiv:2310.15213* (2023).

[28] Turpin, M., Michael, J., Perez, E., and Bowman, S. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. *Advances in Neural Information Processing Systems 36* (2023), 74952–74965.

[29] Tutek, M., Chaleshtori, F. H., Marasović, A., and Belinkov, Y. Measuring chain of thought faithfulness by unlearning reasoning steps. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing* (2025), pp. 9946–9971.

[30] Wang, K., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the wild: a circuit for indirect object identification in gpt-2 small. *arXiv preprint arXiv:2211.00593* (2022).

[31] Wang, S., Zhu, Y., Liu, H., Zheng, Z., Chen, C., and Li, J. Knowledge editing for large language models: A survey. *ACM Computing Surveys 57*, 3 (2024), 1–37.

[32] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q. V., Zhou, D., et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems 35* (2022), 24824–24837.

[33] Wu, C., Wang, X., Lian, D., Xie, X., and Chen, E. A causality inspired framework for model interpretation. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (2023), pp. 2731–2741.

[34] Yang, W., Sun, F., Tan, J., Ma, X., Cao, Q., Yin, D., Shen, H., and Cheng, X. The mirage of model editing: Revisiting evaluation in the wild. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2025), pp. 15336–15354.

[35] Yang, Z., Qi, P., Zhang, S., Bengio, Y., Cohen, W., Salakhutdinov, R., and Manning, C. D. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 conference on empirical methods in natural language processing* (2018), pp. 2369–2380.

[36] Yuan, Y., Zhao, L., Zhang, K., Zheng, G., and Liu, Q. Do llms overcome shortcut learning? an evaluation of shortcut challenges in large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing* (2024), pp. 12188–12200.

[37] Zhu, K., Tu, S., Jin, Z., Hou, L., Li, J., and Zhao, J. Establishing trustworthy llm evaluation via shortcut neuron analysis. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2025), pp. 3809–3822.